Gauge Alignment and Real-Space Locality in Higher-Rank Periodic One-Dimensional Hamiltonians

Analytic controls, numerical methodology, and evidence boundaries

Project Koios

Pre-execution technical report

Abstract

Matrix-valued tight-binding representations of an isolated composite subspace are not unique: a momentum-dependent unitary change of retained basis preserves the represented spectrum while changing individual Fourier hopping blocks and, consequently, the error incurred by finite-range truncation. This report develops a controlled periodic one-dimensional study of that distinction for retained ranks $r \in \{3, 4, 8, 16\}$ and noncommuting gauge-circuit depths $L \in \{1, 2, 4, 8\}$. The parent Hamiltonian, isolated retained subspace, and gauge attack are analytic. Four prescribed retained-space gauge alignments—identity, global-unitary, pointwise-unitary, and known-inverse-oracle—are compared using a common closed-loop transport of the analytic reference frame. Odd construction meshes support finite-periodic matrix Fourier transforms, while a disjoint shifted mesh separates interpolation error from deliberate finite-support truncation error. The design retains rank-normalized block tails, Parseval and Hermiticity defects, operator and ordered-spectrum errors, perturbation bounds, and mandatory mesh-refinement evidence.

Several conclusions follow analytically from the construction: the retained subspace has an exact uniform exterior-isolation energy $2.10E_{\mathrm{ref}}$; the gauge attack preserves the retained projector and spectrum; pointwise-unitary and known-inverse-oracle alignment recover the diagonal reference representation in exact arithmetic; and that recovered field has Fourier support only for $|R| \leq 2$. These are properties by construction, not new numerical findings. The maintained producer, typed evidence records, and a bounded producer-independent reference analysis of analytic-model and attack core invariants, together with bounded direct-sum identity and known-inverse-oracle Fourier controls, are implemented and software-tested. Complete protocol-wide reference analysis, byte-owning verification, resource-qualified execution, and a retained scientific result do not exist. Accordingly, this report presents methodology, analytic results, and software-verification status without reporting unexecuted numerical trends.

Keywords: composite bands; gauge covariance; Procrustes alignment; parallel transport; matrix Fourier transform; hopping locality; truncation error; scientific evidence

1 Introduction

Localized real-space Hamiltonians are routinely obtained by Fourier transforming a matrix-valued reciprocal-space representation. For an isolated composite band subspace, the represented eigenvalues are invariant under momentum-dependent unitary rotations of the retained basis, but the matrix entries and their Fourier coefficients are not. This familiar gauge freedom underlies both the usefulness and the interpretive difficulty of Wannier and tight-binding representations [1, 2, 3]. A short-ranged representation can be advantageous for interpolation and reduced modeling [4], yet

spectral equivalence of the untruncated field alone does not imply that two gauges respond equally to a common finite-range truncation.

The present study isolates this issue from material-specific electronic-structure uncertainties. It does not run Quantum ESPRESSO, Wannier90, or a spread minimization. It instead supplies an analytic, finite-dimensional parent Hamiltonian with an exactly isolated retained subspace and then applies a deterministic higher-rank gauge circuit. The attacked representation has the same pointwise spectrum and projector as the reference representation. Any difference in its hopping locality is therefore a representation effect rather than a change of the underlying retained operator.

The central scientific question is:

> For a fixed isolated retained operator, how do retained rank, noncommuting gauge-circuit depth, and a prospectively selected gauge-alignment family affect matrix-valued hopping locality and the error of deliberate finite-support truncation?

This question is intentionally narrower than the problem of finding optimally localized Wannier functions. The study compares four declared feasible alignment families rather than searching an unrestricted localization functional. In particular, the analytic known inverse is retained only as an oracle control. It cannot support a claim about practical recovery from unknown data.

1.1 Contributions and organization

The report makes five methodological contributions.

(i) It defines a rank-generic analytic family with an exact, rank-independent exterior- isolation certificate and a connected noncommuting gauge attack.

(ii) It distinguishes common reference-frame transport from retained-space gauge alignment and fixes the left/right action conventions needed to compare them.

(iii) It separates identity, global-unitary, pointwise-unitary, and known-inverse-oracle gauge alignments from interpolation, truncation, and total *error channels*.

(iv) It defines an odd-mesh finite-periodic Fourier and support analysis with a disjoint shifted evaluation mesh and explicit mesh-refinement gates.

(v) It binds the scientific stages to immutable typed requests and fail-closed evidence records, while keeping independent recomputation and publication outside the producer.

Section 2 summarizes the scientific background. Sections 3–8 define the methodology. Section 9 states the analytic and implementation results currently established. Section 10 discusses their interpretation and limitations. Section 11 concludes. Detailed numerical controls and record ownership appear in the appendices.

1.2 Evidence status

This is a pre-execution technical report. The authoritative protocol fixes the coefficients, meshes, tolerances, schemas, resource limits, and future evidence policy. The maintained direct runner is explicitly development-only and writes no retained scientific evidence. No numerical trend across the 16 rank/depth cases is asserted here. Passing tests establishes software behavior only; it does not establish physical validity, independent numerical verification, resource compliance, or convergence of a retained calculation.

2 Background and scientific setting

2.1 Represented operators and gauge covariance

Let $\mathcal{H}_{\mathrm{amb}}$ be a finite ambient state space and let $F(k) : \mathbb{C}^r \to \mathcal{H}_{\mathrm{amb}}$ be an orthonormal frame for a rank-r retained subspace at reciprocal coordinate k. For an ambient Hermitian operator $H_p(k)$, the represented retained operator is

$$H_F(k) = F(k)^\dagger H_p(k) F(k). \tag{1}$$

A retained-coordinate gauge transformation acts from the right, $F(k) \mapsto F(k)U(k)$ with $U(k) \in U(r)$, and therefore

$$H_F(k) \mapsto H_{FU}(k) = U(k)^\dagger H_F(k) U(k). \tag{2}$$

Equation (2) preserves the unordered pointwise spectrum and the ambient projector $P(k) = F(k)F(k)^\dagger$. It does not preserve individual matrix entries or Fourier coefficients.

This distinction is fundamental. A physical model is identified by its state spaces, operator, geometry, units, and energy reference. A finite matrix is a representation of that operator in a declared basis. Comparing matrices without first aligning their state spaces and basis conventions can confound a change of representation with a change of model.

2.2 Alignment versus localization

The orthogonal or unitary Procrustes problem selects the unitary factor that best aligns two frame matrices in a Frobenius norm [5]. Its polar factor can be computed from an SVD or polar decomposition [6, 7]. Pointwise Procrustes alignment supplies one unitary per reciprocal point. A global-unitary restriction instead supplies one matrix for the entire reciprocal route. These are different feasible sets, not different numerical accuracies for the same optimization problem.

Neither alignment is equivalent to minimizing a Wannier spread functional. The present protocol therefore does not claim to reproduce maximally localized Wannier functions. Its purpose is diagnostic: it asks how much locality is lost under a known gauge attack and how much is recovered under precisely declared alignment information.

2.3 Fourier locality and finite support

For a periodic represented field $H(k)$, continuum hopping blocks are

$$T(R) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-ikR} H(k)\, dk, \qquad R \in \mathbb{Z}. \tag{3}$$

Rapid decay of $\|T(R)\|$ supports accurate finite-range approximation. Analytic periodicity often implies rapid Fourier decay, but a nonlinear momentum-dependent gauge rotation can generate an infinite harmonic inventory even when the reference field has finite support. Thus "same bands" and "same truncation behavior" are distinct statements.

A sampled finite transform introduces two additional distinctions. First, discrete coefficients are a finite representation of sampled data and may contain aliasing from unresolved continuum harmonics. Second, deliberately setting blocks outside a chosen support to zero is a model-reduction operation. The protocol keeps continuum behavior, finite-mesh interpolation, and deliberate truncation separate.

3 Analytic model

3.1 Geometry, state spaces, and units

The selected D1 specialization uses the dimensionless unit lattice and reduced Brillouin torus $\mathbb{T} = [-\pi, \pi)$. For retained rank $r \geq 3$, the ambient state space is $\mathbb{C}^{2r}$ with the retained coordinates ordered before the exterior coordinates. The canonical embedding $E_r : \mathbb{C}^r \to \mathbb{C}^{2r}$ and projector are

$$E_r = \begin{pmatrix} I_r \\ 0 \end{pmatrix}, \qquad P_0 = E_r E_r^\dagger. \tag{4}$$

All energies are expressed in a declared dimensionless unit with positive reference scale $E_{\mathrm{ref}} = 1$ during numerical evaluation. Formulas retain E_{ref} to expose the normalization.

The P0 control uses the constant analytic frame $F_r(k) = E_r$ and identity ambient sewing. The state-space map is therefore known directly rather than inferred from eigenvectors. Internal crossings among retained energies are permitted because the complete retained projector, not a sequence of individually labeled bands, is the scientific object.

3.2 Rank-normalized parent spectrum

For retained coordinate $a \in \{1, \ldots, r\}$, define

$$x_{r,a} = \frac{2a - r - 1}{r - 1} \in [-1, 1], \qquad c(k) = \frac{3}{4}\cos k + \frac{1}{4}\cos 2k. \tag{5}$$

The retained and exterior energies are

$$\frac{\epsilon_a(k)}{E_{\mathrm{ref}}} = 0.40 x_{r,a} + 0.25\,(1 + 0.20 x_{r,a})\,c(k), \tag{6}$$

$$\frac{\epsilon_a^{\mathrm{ext}}(k)}{E_{\mathrm{ref}}} = 3.00 + 0.30 x_{r,a} + 0.10 c(k). \tag{7}$$

Let $D_r(k) = \mathrm{diag}(\epsilon_1(k), \ldots, \epsilon_r(k))$ and define the ambient parent

$$H_p^{(r)}(k) = \begin{pmatrix} D_r(k) & 0 \\ 0 & D_{\mathrm{ext}}(k) \end{pmatrix}. \tag{8}$$

The represented reference operator is exactly $F_r^\dagger H_p^{(r)} F_r = D_r$.

Proposition 1 (Exact exterior isolation)**.** *For every selected finite rank and every $k \in \mathbb{T}$, the exterior spectrum lies at least $2.10 E_{\mathrm{ref}}$ above the retained spectrum. The lower bound is attained at $k = 0$.*

Proof. For fixed k, the retained maximum occurs at $x = 1$ and the exterior minimum at $x = -1$. Using Equations (6)–(7), their separation is $(2.30 - 0.20 c(k)) E_{\mathrm{ref}}$. Since $|c(k)| \leq 1$, the separation is at least $2.10 E_{\mathrm{ref}}$, and $c(0) = 1$ gives equality. □

The proposition is an analytic property of the finite model. A sampled minimum is retained separately as a numerical reconstruction check and is never relabeled as the continuum certificate.

3.3 Deterministic noncommuting gauge attack

Adjacent retained coordinates are mixed by real Givens rotations. For layer ℓ and pair start j, define

$$y_{r,j} = \frac{2j - r}{r - 1}, \qquad s_\ell(k) = \frac{3}{4}\sin k + \frac{1}{4}(-1)^\ell \sin 2k, \tag{9}$$

$$\theta_{\ell j}(k) = 0.35(-1)^{\ell+j}\,(1 + 0.15 y_{r,j})\, s_\ell(k). \tag{10}$$

Layer ℓ acts on the disjoint adjacent pairs with $j \equiv \ell \pmod 2$. Factors within one layer commute; neighboring layers generally do not. Increasing layer order is part of the definition. The depth-L product is denoted $G_{r,L}(k) \in SO(r)$.

The attacked frame and represented operator are

$$F'_{r,L}(k) = F_r(k) G_{r,L}(k), \qquad H'_{r,L}(k) = G_{r,L}(k)^\dagger D_r(k) G_{r,L}(k). \tag{11}$$

One layer is an incomplete-connectivity baseline; two or more alternating layers cover every adjacent edge and form a connected chain. Increasing depth changes the representation while leaving the retained operator and projector invariant.

Proposition 2 (Exact covariant equivalence)**.** *For every r, L, and k, the attacked frame is orthonormal, its ambient projector is P_0, and $H'_{r,L}(k)$ has the same unordered spectrum as $D_r(k)$.*

Proof. Each Givens factor is orthogonal, hence $G_{r,L}^\dagger G_{r,L} = I_r$. Substitution into Equation (11) proves frame orthonormality and projector equality. The represented operators are related by a unitary similarity, which proves spectral equivalence. □

The nonlinear entries of $G_{r,L}$ generally contain harmonics beyond those present in the angle functions. Consequently the attacked represented field is analytic and periodic but is not assigned a finite Fourier support by construction.

4 Gauge-alignment methodology

4.1 Canonical source and target orientation

Let $R(k) = F_r(k)$ be the reference, $C(k) = F'_{r,L}(k)$ the attacked candidate, and $H_C(k) = H'_{r,L}(k)$ the represented candidate operator. Every prescribed gauge alignment acts from the right:

$$\widehat{F}_U(k) = C(k)U(k), \qquad \widehat{H}_U(k) = U(k)^\dagger H_C(k) U(k). \tag{12}$$

Returning $U^\dagger$ while retaining Equation (12) would be a convention error, not an equivalent implementation.

For cross-covariance $X = C^\dagger R$ with complete SVD $X = L\Sigma V^\dagger$, the admitted unitary polar factor is

$$\mathcal{P}(X) = LV^\dagger. \tag{13}$$

It is selected only when

$$\sigma_{\min}(X) > a u_{\mathrm{sv}}(X), \qquad \tau_{\mathrm{sv}}(X) = \max(10^{-12}, 10^{-10}\sigma_{\max}(X)). \tag{14}$$

Equality is deficient. No arbitrary null-space completion is introduced.

4.2 Four prescribed gauge alignments

Identity, wire value `identity`. $U_{\mathrm{id}}(k) = I_r$. This control measures the attacked representation as supplied.

Global-unitary, wire value `global-unitary`. One matrix minimizes the equal-weight objective

$$J_{\mathcal{S}}(U) = \frac{1}{|\mathcal{S}|r} \sum_{k \in \mathcal{S}} \|C(k)U - R(k)\|_{\mathrm{F}}^2. \tag{15}$$

The selection sample is exactly the final construction mesh $\mathcal{K}_{2049}$. The selected unitary is reused on the coarser construction meshes and shifted evaluation mesh without refitting.

Pointwise-unitary, wire value `pointwise-unitary`. At every construction point, $U_{\mathrm{pt}}(k) = \mathcal{P}(C(k)^\dagger R(k))$. Selection is repeated separately on each construction mesh. Selected pointwise unitaries are not interpolated to the shifted evaluation mesh; the direct recovered reference there is the analytic $D_r(k)$.

Known-inverse-oracle, wire value `known-inverse-oracle`. $U_{\mathrm{oracle}}(k) = G_{r,L}(k)^{-1} = G_{r,L}(k)^\dagger$. This action performs no optimization. It checks construction, orientation, application, and Fourier conventions. It is not a practical estimator.

The four names identify gauge alignments, not diagnostic channels. "Error channel" is reserved for the later interpolation, truncation, and total residual decomposition.

4.3 Alignment integrity evidence

For every selected Procrustes factor, the producer retains all singular values, the applied cutoff, strict rank margin, condition number when defined, and the objective recomputed from the original frames. It also evaluates

$$\varepsilon_{\mathrm{unit}} = \|U^\dagger U - I_r\|_{\mathrm{F}}, \qquad \varepsilon_{\mathrm{stat}} = \frac{\|U^\dagger X - X^\dagger U\|_{\mathrm{F}}}{\sqrt{r}}, \tag{16}$$

plus the covariance-objective identity and the negative part of the Hermitian polar factor. These checks distinguish a nominally returned matrix from a scientifically admitted alignment.

Proposition 3 (Pointwise and oracle recovery)**.** *For the P0 construction, pointwise-unitary and known-inverse-oracle alignment both recover $F_r(k)$ and $D_r(k)$ in exact arithmetic.*

Proof. Here $C(k) = F_r G_{r,L}$ and $R(k) = F_r$, so $C(k)^\dagger R(k) = G_{r,L}(k)^\dagger$. This matrix is unitary and has all singular values equal to one. Its unique full polar factor is itself, which equals the known inverse. Substitution into Equation (12) gives the claim. □

The two actions remain separate in software because one reconstructs a factor from sampled frames while the other receives the generating inverse. Their agreement tests an implementation contract; it is not an empirical discovery.

5 Common closed-loop frame transport

Transport is applied to the analytic reference frame and is common to all four gauge alignments. It is not a fifth alignment and is not appended only to the pointwise case. This separation prevents transport defects from being misattributed to gauge-selection quality.

On the ordered construction mesh, let $Y_n = F_r(k_n)$ and set $\widetilde{F}_0 = Y_0$. At each neighbor link, form

$$O_n = \widetilde{F}_n^\dagger Y_{n+1} = L_n \Sigma_n R_n^\dagger, \qquad Q_n = R_n L_n^\dagger, \qquad \widetilde{F}_{n+1} = Y_{n+1} Q_n. \tag{17}$$

Then $\widetilde{F}_n^\dagger \widetilde{F}_{n+1}$ is Hermitian positive semidefinite. The strict rank rule in Equation (14) applies to every link. A deficient link terminates dependent transport; later links are unattempted rather than fabricated.

After traversing the period, write the endpoint mismatch as

$$\mathcal{H} = Z \operatorname{diag}(e^{i\gamma_1}, \ldots, e^{i\gamma_r}) Z^\dagger, \qquad \gamma_a \in (-\pi, \pi]. \tag{18}$$

The branch choice is part of the method. It assigns $+\pi$ to eigenvalue -1 and thereby fixes the distributed root. Closed frames are

$$F_n^{\text{closed}} = \widetilde{F}_n Z \operatorname{diag}(e^{-in\gamma_1/M}, \ldots, e^{-in\gamma_r/M}) Z^\dagger. \tag{19}$$

If $\widetilde{F}_M = \widetilde{F}_0 \mathcal{H}$, then $F_M^{\text{closed}} = F_0^{\text{closed}}$. For the constant P0 reference frame and identity sewing, the exact model predicts trivial transport and zero endpoint phases; the numerical transport remains valuable as an action-orientation and closure control.

The phase tuple is an unordered multiset scientifically. Deterministic sorting is a wire convention, not a continuation of individual band labels through degeneracies.

6 Finite-periodic Fourier and locality methodology

6.1 Construction and evaluation meshes

The three construction meshes are

$$\mathcal{K}_M = \left\{ -\pi + \frac{2\pi n}{M} : n = 0, \ldots, M-1 \right\}, \qquad M \in \{513, 1025, 2049\}. \tag{20}$$

All are odd, centered, and half open. The direct evaluation mesh is

$$\mathcal{K}_{\text{eval}} = \left\{ -\pi + \frac{2\pi(m+1/2)}{8191} : m = 0, \ldots, 8190 \right\}. \tag{21}$$

It is disjoint from every construction mesh and cannot influence a fit, support, tolerance, or reporting rule.

For odd M, the canonical translation inventory is $\mathcal{R}_M = \{-(M-1)/2, \ldots, (M-1)/2\}$. There is no self-inverse Nyquist singleton; the canonical disposition is `absent-on-odd-mesh`.

6.2 Discrete transform and support policy

For one gauge alignment a, the finite-periodic transform is

$$T_{a,M}(R) = \frac{1}{M} \sum_{k \in \mathcal{K}_M} e^{-ikR} H_a(k), \qquad H_{a,M}(k) = \sum_{R \in \mathcal{R}_M} e^{ikR} T_{a,M}(R). \tag{22}$$

Only the forward transform carries $1/M$. The complete coefficient inventory is used for reconstruction and tail calculations. The public result retains the bounded norm inventory $|R| \le 128$ rather than serializing all dense coefficient arrays.

The frozen supports are

$$\Omega_S = \{R \in \mathbb{Z} : |R| \le S\}, \qquad S \in \{0, 1, 2, 4, 8, 16, 32, 64, 128\}. \tag{23}$$

Every support contains inversion pairs. All nine are evaluated; no first-passing or plot-selected support is promoted.

For the reference diagonal field, Equations (6) and (3) imply an exact support certificate.

Proposition 4 (Reference support)**.** *The continuum Fourier coefficients of $D_r(k)$ vanish for every $|R| > 2$.*

Proof. Every diagonal entry of D_r is a linear combination of 1, $\cos k$, and $\cos 2k$. These harmonics have Fourier indices only at 0, ± 1, and ± 2. □

Corollary 1. *In exact arithmetic, pointwise-unitary and known-inverse-oracle alignment recover a represented field with continuum support $|R| \le 2$.*

This is an analytic certificate. Finite-transform reconstruction and binary64 residuals still require numerical checks.

6.3 Tail and error diagnostics

For support Ω_S, define the omitted quadratic weight, its square root, and the block-ℓ^1 tail:

$$W_{a,M}(S) = \sum_{R \notin \Omega_S} \|T_{a,M}(R)\|_{\mathrm{F}}^2, \tag{24}$$

$$N_{a,M}(S) = \sqrt{W_{a,M}(S)}, \tag{25}$$

$$L_{a,M}(S) = \sum_{R \notin \Omega_S} \|T_{a,M}(R)\|_{\mathrm{F}}. \tag{26}$$

Cross-rank comparisons retain $W/(rE_{\mathrm{ref}}^2)$ and $N/(\sqrt{r}E_{\mathrm{ref}})$ in addition to raw values. This prospective normalization prevents matrix dimension alone from determining a trend.

On the shifted mesh, the full interpolant, truncated interpolant, and direct analytic reference define

$$e_{\mathrm{interp}}(k) = H_{a,M}(k) - H_a^{\mathrm{direct}}(k), \tag{27}$$

$$e_{\mathrm{trunc}}^{(S)}(k) = H_{a,M}^{(S)}(k) - H_{a,M}(k), \tag{28}$$

$$e_{\mathrm{total}}^{(S)}(k) = H_{a,M}^{(S)}(k) - H_a^{\mathrm{direct}}(k). \tag{29}$$

The pointwise identity $e_{\mathrm{total}} = e_{\mathrm{interp}} + e_{\mathrm{trunc}}$ is checked directly. Scalar maxima of the three error channels are not added: their maximizing points can differ, and cancellation in the total error is reported rather than called convergence.

For each residual, the study records maximum and RMS Frobenius operator error and maximum and RMS ordered-spectrum error. Ordered spectra are compared as sorted multisets rather than by tracking eigenvectors through crossings. At every point, the finite-dimensional bounds

$$\max_i |\lambda_i(H_{\mathrm{approx}}) - \lambda_i(H_{\mathrm{reference}})| \le \|e\|_2 \le \|e\|_{\mathrm{F}} \tag{30}$$

are checked [8]. Qualified saturation ratios are reported only when their denominators exceed the frozen floor.

7 Numerical study design

The complete matrix contains 16 rank/depth cases, three construction meshes, four gauge alignments, and nine supports. Before failure dispositions, this corresponds to 192 mesh/alignment combinations and 1728 support rows. The global-unitary fit occurs once per case at $M = 2049$ and is reused; it is not counted as a new fit on coarser or evaluation meshes.

Figure 1 summarizes the scientific flow. Solid arrows denote fields used by later calculations. Dashed arrows denote common transport evidence that is correlated into the aggregate result but does not redefine a gauge alignment.

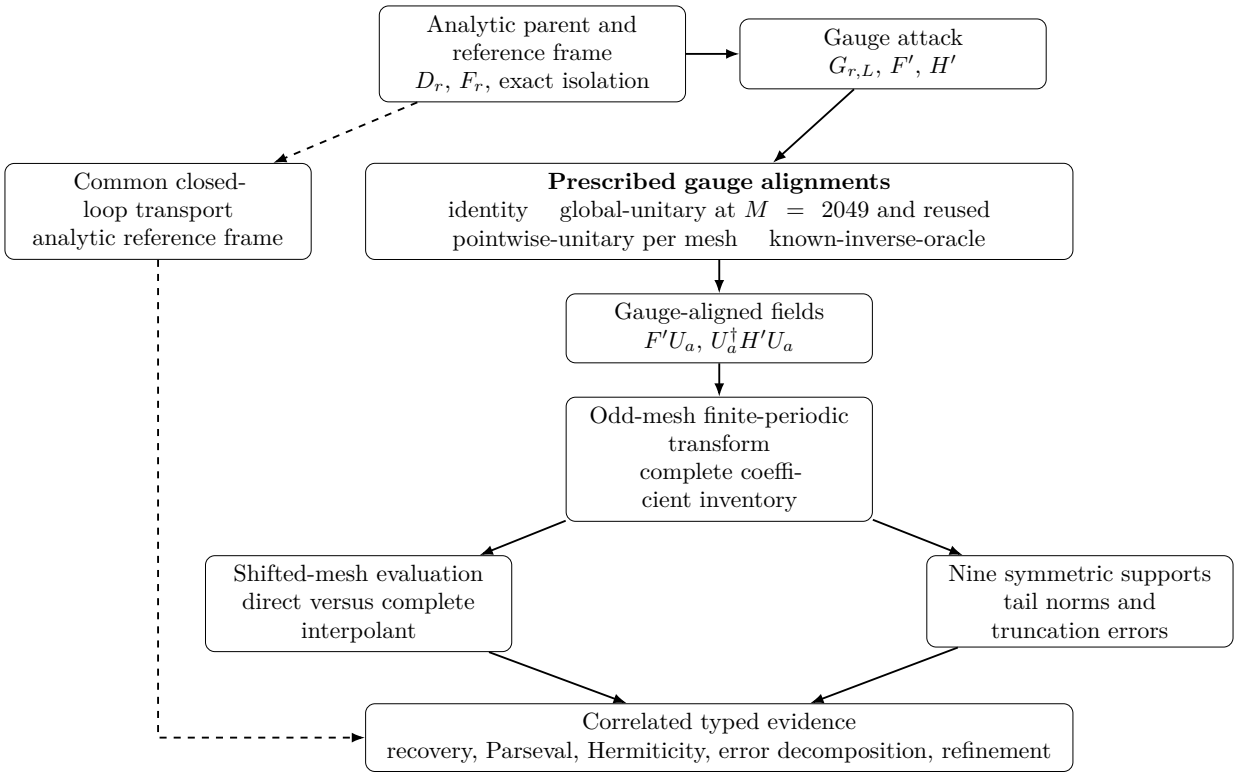


Figure 1: Scientific flow of one higher-rank case. Gauge alignment is performed before Fourier transformation and deliberate truncation.

7.1 Refinement and qualification

The smaller meshes are mandatory refinement controls. Blockwise comparisons cover both adjacent mesh pairs. The final scalar inventory uses the mandatory $1025 \to 2049$ pair and the absolute-relative rule

$$|q_f - q_c| \le \tau_{\mathrm{abs}} + \tau_{\mathrm{rel}} \max(|q_c|, |q_f|), \tag{31}$$

with a denominator floor for the separately reported relative change. The shared `ScalarRefinementAnalyzer` owns this arithmetic; the higher-rank diagnostic inventory owner defines which scientific quantities enter the ledger.

A case is numerically qualified only if its mandatory construction, model, transport, alignment, Fourier, interpolation, and refinement gates pass. A global-unitary aggregate may instead carry the bounded disposition `aggregate-rank-deficient`; this makes only global-unitary downstream

values inapplicable. Pointwise rank deficiency, documented backend failure, alignment integrity failure, or oracle integrity failure invalidates the complete case.

7.2 Claim-resolution rules

The protocol does not infer a locality effect from visual separation alone. For a qualified case, gauge-induced locality extension at $S = 2$ is resolved only if the identity tail exceeds the frozen 10^{-6} scale, the oracle tail is below 10^{-10}, and the identity ordered-spectrum truncation error also exceeds the effect-resolution floor. Rank or depth differences are descriptive unless their normalized difference exceeds both 10^{-6} and ten times the larger observed refinement change. These rules are fixed before execution and do not assert that an effect will be found.

No monotonic trend with rank or depth is assumed. Noncommuting products can produce cancellation, and a finite ladder cannot establish an asymptotic law.

8 Diagnostic and evidence methodology

8.1 Metric-explicit diagnostics

Every diagnostic names its norm and normalization. Frame and unitary residuals use $\|\cdot\|_{\mathrm{F}}/\sqrt{r}$ when rank-normalized; operator residuals use $\|\cdot\|_{\mathrm{F}}/(\sqrt{r}E_{\mathrm{ref}})$. Maximum-absolute-entry defects used internally by a backend are not relabeled as Frobenius diagnostics. Raw and normalized quantities remain available for different scientific questions.

The transform checks include representative count, uniqueness, ordering, inversion closure, reconstruction, Hermitian pairing, Parseval equality, nested-support monotonicity, and the explicit odd-mesh Nyquist disposition. These are integrity checks, not alternative definitions of locality.

8.2 Fail-closed numerical outcomes

Rank deficiency, integrity failure, and backend failure have different meanings and remain distinct typed outcomes. Documented backend failures are converted into stable codes for nonfinite arithmetic, nonconvergence, or validation failure. Runtime exception messages, tracebacks, host paths, and fabricated numerical placeholders are excluded. Downstream values that were not scientifically established are absent, not zero.

8.3 Typed stage ownership

The maintained producer decomposes one gauge-alignment analysis into recovery, Fourier, interpolation, locality, and assembly stages. Each stage binds a complete immutable request and exposes a zero-argument execution. Stage evidence retains its exact source request. Assembly rejects same-shaped but uncorrelated arrays, a replacement request, wrong mesh coordinates, mismatched case identity, wrong gauge alignment, or inconsistent selection provenance.

Stage owner	Scientific responsibility	Evidence boundary
`fields.py`	Construct one immutable represented operator and frame field.	No parallel uncorrelated dictionaries.
`recovery.py`	Compare fields with analytic or solver-owned references.	Frame and operator recovery remain separate.
`fourier.py`	Apply the PhysKit odd-mesh transform and structural checks.	Coefficients are not silently normalized or reindexed.
`interpolation.py`	Evaluate complete Fourier series and direct fields on the shifted mesh.	Evaluation coordinates are immutable and disjoint.
`locality.py`	Evaluate every support and the three error channels.	No preferred support is selected.
`assembly.py`	Correlate stages and construct the public mesh result.	No scientific recomputation during assembly.

Canonical typed JSON bytes and SHA-256 digests establish record identity and provenance. They do not establish scientific correctness. Independent recomputation must bypass the producer, and rendering must consume authenticated values without calculating new scientific quantities.

9 Results established before retained execution

9.1 Analytic results

Table 1 collects the results proved from the selected formulas. They require no numerical execution and must not be reported as empirical observations.

Result	Statement	Interpretive boundary
Exterior isolation	$\Delta_{\mathrm{iso}} = 2.10 E_{\mathrm{ref}}$ for every selected rank.	Exact finite-model certificate; not a material band gap.
Gauge covariance	Attack preserves projector and pointwise spectrum.	Does not preserve matrix entries or hopping locality.
Pointwise recovery	Pointwise polar factor equals $G^{\dagger}_{r,L}$.	Exact P0 property; not evidence for noisy or unknown frames.
Oracle recovery	Known inverse restores F_r and D_r.	Construction control, not a fitted solution.
Reference support	$T_{D_r}(R) = 0$ for $\lvert R\rvert > 2$.	Continuum statement; finite-transform defects remain numerical.
Hermitian pairing	$T(-R) = T(R)^{\dagger}$.	Exact continuum identity tested separately on each finite mesh.

Table 1: Analytic results available before numerical execution.

The exact results establish a controlled contrast. Identity and global-unitary alignment are not assigned finite support by construction, while pointwise-unitary and oracle alignment recover a field whose exact support is known. Whether the selected finite attack produces a *resolved* tail or truncation effect at each (r, L) remains a numerical question governed by the frozen thresholds.

9.2 Implementation and software-verification results

The maintained Python producer implements the analytic family, global and pointwise Procrustes selection, common PhysKit transport, odd-mesh finite-periodic transforms, shifted interpolation,

support diagnostics, refinement, immutable typed records, and canonical serialization. The former monolithic gauge-alignment analysis has been split into cohesive request-bound scientific owners. A separate bounded reference action now recomputes identity and known-inverse-oracle Fourier blocks by direct sums on every construction mesh. It compares reconstruction, Parseval and Hermitian pairing, retained block norms, shifted interpolation, and support tails for both controls, together with the oracle's analytic $|R| \leq 2$ coefficients, without invoking producer analysis actions. Gauge-alignment wire values are descriptive and strictly mapped from a plain domain enum; obsolete opaque values and the former ambiguous key are rejected.

At the time of this report revision, the focused higher-rank package suite passes 68 tests, the calculation-local structural suite passes 4 tests, and the complete bounded package suite passes 4705 tests with 23 environment- or historical-tree-dependent skips and 89 explicitly expensive deselections. Ruff, strict source mypy, strict Sphinx, `compileall`, LaTeX builds, and `git diff -check` also pass. These results establish current software behavior and document buildability only. They do not constitute independent numerical verification of a scientific result.

9.3 Numerical results not yet available

No complete 16-case development execution was performed for this report revision. No retained result bytes have been admitted. The bounded references do not cover transport, Procrustes, global- or pointwise-unitary Fourier values, support truncation and total-error fields, error-channel decompositions, or refinement. The 4 GiB resource ceiling has not been demonstrated. Consequently this report does not provide rank or depth trends, support Pareto fronts, global-unitary residuals, wall-time scaling, or retained figures. Empty result space is reported explicitly rather than populated with illustrative or synthetic values that could be mistaken for calculated evidence.

A later result section may be populated only from a separately authorized retained execution followed by independent recomputation and byte-owning verification. Its tables must distinguish calculated values from analytic certificates and must preserve unresolved or inapplicable dispositions.

10 Discussion

10.1 What the controlled comparison can establish

The construction isolates representation dependence. Since the parent, projector, and untruncated retained spectrum are held fixed, a resolved difference in finite-support error would demonstrate that short-range compressibility depends on retained-space gauge within this finite analytic family. It would not demonstrate that the parent model is physically realistic or that one gauge is universally optimal.

Pointwise-unitary and oracle recovery form a strong implementation control because their exact target is known. Global-unitary alignment is scientifically different: one matrix cannot generally undo a momentum-dependent noncommuting attack. A nonzero qualified global residual would measure the limitation of that restricted feasible set, not solver failure. Conversely, a small residual at one finite case would not imply equivalence of global and pointwise alignment in general.

10.2 Interpreting rank and depth

Retained rank changes matrix dimension and the number of adjacent mixing edges. Circuit depth changes the ordered product of generally noncommuting rotations. The selected $\sqrt{r}$ and r normalizations make cross-rank magnitudes interpretable without removing all rank dependence. Raw values remain necessary because normalization itself answers a different question.

Depth is not a scalar attack amplitude. Additional layers can reinforce or cancel existing harmonics, so locality need not vary monotonically with L. The finite set $L \in \{1, 2, 4, 8\}$ supports controlled comparisons only; it does not define a continuum or asymptotic-depth limit.

10.3 Transport and topological scope

The common closed-loop transport tests numerical continuity, endpoint sewing, and branch conventions. In P0 the exact reference projector is constant and topologically trivial. Closure phases are therefore transport diagnostics, not polarization, Wannier centers, or a topological invariant. The distributed closure is a smooth-gauge construction, not a localization optimum.

A moving-projector or topologically nontrivial study would require patching, sewing, and obstruction contracts not present here. Such a study cannot be inferred by replacing the constant frame with backend eigenvectors while retaining the present evidence semantics.

10.4 Higher dimensions and nonorthonormal bases

Fiberwise equations such as $F'U$ and $U^\dagger H'U$ do not intrinsically select a spatial dimension. Meshes, Fourier indices, inversion closure, and support geometry do. The present Fourier and locality actions are therefore deliberately one-dimensional. A 2-D or 3-D study must define reviewed tensor or nonorthogonal reciprocal meshes, multi-index representatives, complete point-group or inversion orbits, and a physical support metric. No placeholder implementation is supplied.

Likewise, the current frames are orthonormal and their gauge group is $U(r)$. A nonorthonormal basis requires a positive-definite fiber metric, metric adjoint, generalized isometry group, conditioning evidence, and a metric-aware Procrustes objective. Treating that change as an optional parameter of the present action would hide a new mathematical contract.

10.5 Reproducibility and evidence lifecycle

The intended lifecycle is calculate $\rightarrow$ independently recompute $\rightarrow$ verify exact bytes $\rightarrow$ render authenticated values. The producer owns scientific calculation but not its independent reference. The verifier owns the admitted bytes and compares them with independently reconstructed formulas. The renderer owns presentation only.

The calculation-local `workflow/` directory is historical provenance and remains outside the maintained runtime and tests. The direct development runner composes the maintained owners explicitly and labels its summary `scientific_evidence: false`. No result is sealed, published, or released by executing that runner.

10.6 Limitations

The principal limitations are:

1. the parent is synthetic, finite dimensional, and dimensionless;
2. only the constant-projector real-gauge P0/G0 specialization is implemented;
3. the oracle assumes knowledge unavailable in an ordinary material calculation;
4. only bounded core-invariant and identity/oracle Fourier reference analyzers are implemented; complete protocol-wide independent analysis and byte-owning verification are absent;
5. complete runtime and memory compliance are unestablished;

6. the finite rank/depth ladder supports no asymptotic law;

7. no 2-D, 3-D, nonorthonormal, moving-subspace, or topological result is available; and

8. passing tests does not establish scientific validation.

11 Conclusion

This report defines a publication-grade methodological account of a controlled higher-rank gauge-locality study without overstating its evidence status. The analytic family cleanly separates a physical retained operator from its finite matrix representation. A deterministic noncommuting gauge circuit changes matrix-valued hopping locality while preserving the retained projector and pointwise spectrum. Four prospectively fixed gauge alignments probe different information constraints, and a common transport control prevents frame-sewing effects from being conflated with gauge selection.

The exact construction already establishes a uniform isolation energy, covariant spectral equivalence, pointwise and oracle recovery, and finite reference support. The software producer implements the planned finite numerical analysis with typed immutable evidence and fail-closed outcomes. The substantive numerical questions—how locality varies with rank and depth, how much a single global unitary can recover, and which finite supports are adequate under withheld evaluation—remain unanswered until an authorized execution is independently verified.

The next scientific gate is therefore not publication but completion of the remaining independent reference analysis and byte-owning verification. Only after those owners, resource-qualified execution, and retained evidence review exist can numerical results be inserted into the report and considered for publication.

A Frozen numerical inventory

Quantity	Frozen value
Retained ranks	$3, 4, 8, 16$
Circuit depths	$1, 2, 4, 8$
Ambient dimension	$2r$
Construction meshes	$513, 1025, 2049$ points
Global-unitary selection mesh	2049 points only
Shifted evaluation mesh	8191 points
Gauge alignments	identity; global-unitary; pointwise-unitary; known-inverse-oracle
Supports	$S = 0, 1, 2, 4, 8, 16, 32, 64, 128$ with $\lvert R\rvert \le S$
Singular-value cutoff	$\max(10^{-12}, 10^{-10}\sigma_{\max})$; equality deficient
Principal phase branch	$(-\pi, \pi]$
Exact isolation certificate	$2.10E_{\mathrm{ref}}$
Mandatory final refinement	$1025 \to 2049$
Arithmetic	binary64

B Frozen tolerance crosswalk

The authoritative protocol prevails if this explanatory table ever differs from the selected schema.

Table 2: Frozen D1 P0/G0 numerical tolerances.

Diagnostic	Normalization	Tolerance
momentum and endpoint coordinates	radian	10^{-14}
frame orthonormality, gauge/alignment unitarity, projector idempotency	raw Frobenius	10^{-12}
Procrustes stationarity, objective identity, and negative polar part	dimensionless	10^{-12}
smooth/attacked projector equality	raw Frobenius	10^{-12}
direct ordered-spectrum equivalence	E_{ref}	10^{-11}
pointwise/oracle frame recovery	$\sqrt{r}$	10^{-11}
pointwise/oracle represented-operator recovery	$\sqrt{r}E_{\mathrm{ref}}$	10^{-11}
endpoint sewing and primitive closure	normalized Frobenius	10^{-11}
maximum closure-phase magnitude	radian	10^{-10}
Fourier reconstruction and Hermitian pairing	$\sqrt{r}E_{\mathrm{ref}}$	2×10^{-11}
relative Parseval defect	dimensionless	2×10^{-11}
error decomposition and support monotonicity	dimensionless	2×10^{-11}
maximum perturbation-bound violation	E_{ref}	10^{-11}
final interpolant operator error	$\sqrt{r}E_{\mathrm{ref}}$	10^{-9}
final interpolant ordered-spectrum maximum error	E_{ref}	10^{-10}
retained-block refinement, $\lvert R\rvert \leq 128$	$\sqrt{r}E_{\mathrm{ref}}$ per block	10^{-10}

C Result and failure semantics

Disposition	**Meaning**	**Downstream consequence**
`qualified`	Required scientific and integrity gates pass.	Declared downstream values are present.
`aggregate-rank-deficient`	Global covariance fails the strict rank gate.	Only global-unitary downstream values are absent.
`pointwise-rank-deficient`	At least one pointwise covariance fails the strict gate.	Complete P0 case is invalid.
`alignment-numerically-unresolved`	Documented backend operation fails or becomes nonfinite.	No selected factor or locality values are fabricated.
`alignment-integrity-failed`	Returned factor fails unitarity, stationarity, objective, or PSD checks.	Complete case is invalid.
`oracle-integrity-failed`	Analytic inverse fails construction or recovery checks.	Complete case is invalid.

D Prospective retained-results presentation

A later independently verified result package should populate, at minimum:

1. one qualification table for all 16 cases and four gauge alignments;
2. global and pointwise rank-margin summaries;
3. common transport and endpoint-closure diagnostics;
4. normalized block-norm decay for every declared support;
5. separate interpolation, truncation, and total operator and spectrum errors;

6. the complete $1025 \to 2049$ scalar-refinement ledger;
7. rank/depth comparisons with the frozen resolution rule;
8. resource measurements for producer and independent verifier; and
9. exact result, verification, provenance, and rendering byte identities.

No table should replace inapplicable or unresolved entries with zero. No plot should select a preferred support unless a separately reviewed decision rule is added before execution.

References

[1] Nicola Marzari and David Vanderbilt. Maximally localized generalized Wannier functions for composite energy bands. *Physical Review B*, 56(20):12847–12865, 1997.

[2] Ivo Souza, Nicola Marzari, and David Vanderbilt. Maximally localized wannier functions for entangled energy bands. *Physical Review B*, 65:035109, 2001.

[3] Nicola Marzari, Arash A. Mostofi, Jonathan R. Yates, Ivo Souza, and David Vanderbilt. Maximally localized wannier functions: Theory and applications. *Reviews of Modern Physics*, 84:1419–1475, 2012.

[4] Jonathan R. Yates, Xinjie Wang, David Vanderbilt, and Ivo Souza. Spectral and Fermi surface properties from Wannier interpolation. *Physical Review B*, 75(19):195121, 2007.

[5] Peter H. Schönemann. A generalized solution of the orthogonal Procrustes problem. *Psychometrika*, 31(1):1–10, 1966.

[6] Nicholas J. Higham. Computing the polar decomposition—with applications. *SIAM Journal on Scientific and Statistical Computing*, 7(4):1160–1174, 1986.

[7] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, Baltimore, 4 edition, 2013.

[8] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, Cambridge, 2 edition, 2013.